

Erdős Problem #912

EP-912

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

If $n! = \prod_i p_i^{k_i}$ is the factorisation into distinct primes then let $h(n)$ count the number of distinct exponents k_i .

Prove that there exists some $c > 0$ such that $h(n) \sim c \left(\frac{n}{\log n} \right)^{1/2}$ as $n \rightarrow \infty$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>